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Optimization of CZTSSe solar cells by defect analysis and control to reduce recombination with WS_2 non-toxic buffer layer and BSF structures

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ABSTRACT

Kesterite-based solar cells (CZTSSe) are promising low-cost, low-toxicity alternatives to conventional solar cells. However, their efficiency is often limited by charge carrier recombination and the need for efficient, non-toxic buffer layers to replace commonly used CdS. In this study, a theoretical model of a CZTSSe solar cell was developed using SCAPS-1D simulations to optimize the cell design and enhance efficiency. Initial simulations with a CdS buffer layer aligned with experimental results. Subsequently, WS_2 was explored as a Cd-free buffer layer due to its suitable band gap, high conductivity, and optical transparency. By optimizing the absorber layer's thickness and doping density in the Mo/CZTS/ WS_2/i:ZnO/Al:ZnO configuration, an efficiency of 12.8 % was achieved, mainly limited by carrier recombination at interfaces. To address this limitation, a back surface field (BSF) layer was introduced between the CZTSSe absorber and the metal back contact. A comparative analysis of V_2O_5 and Cu_2O as BSF materials demonstrated significant performance improvements, with efficiencies of 19.77 % and 19.07 %, respectively, attributed to reduced charge recombination. This improvement is attributed to the reduction in charge carrier recombination, as confirmed by carrier generation and recombination analyses. The study further investigates the impact of defect density in the absorber and at interfaces, as well as temperature effects on cell performance. Key parameters such as activation energy, diffusion voltage, and C-V and C-f curve analyses were extracted and analyzed. This research offers substantial progress in optimizing CZTSSe solar cells, advancing the development of efficient and environmentally friendly photovoltaics.

1. Introduction

In recent years, solar cells made from $Cu_2ZnSn(S;Se)_4$ (CZTSSe) kesterite, which is used as an absorber layer for thin solar cells, have gained popularity [1,2]. This is mainly due to the natural richness of the components they contain, as well as their lack of toxicity. This material has beneficial characteristics such as a high absorption coefficient, in excess of $10^4 cm^{-1}$, and a direct band gap of between 1 eV and 1.5 eV [3–5], depending on the sulfur/selenium (S/Se) ratio. These attributes make them an attractive option for solar cells.

Despite encouraging prospects, CZTSSe faces a number of technical constraints. The manufacture of this material comes up against major complications, particularly regarding the exact control of its chemical composition [6,7]. This problem leads to the unwanted appearance of additional compounds such as ZnS, SnSe and Cu_2SnS_3, making it difficult to obtain pure CZTSSe. Another critical point concerns the limited electrical performance of the material, characterized by poor charge transport. Structural imperfections, particularly antisite defects, such as $Cu_{(Zn)}$, $Zn_{(Cu)}$, $Cu_{(Sn)}$, $Sn_{(Cu)}$, $Zn_{(Sn)}$ and $Sn_{(Zn)}$ [5,8,9]. These defects favor the recombination of charge carriers, which negatively affects the open circuit voltage (V_{oc}) and compromises the overall efficiency of photovoltaic devices using this material [10]. The integration of CZTSSe as an absorber material in thin-film photovoltaic devices is performing below expectations. In 2021, Qingbo Meng and his team achieved a record efficiency of 13.5 % using an architecture with a CdS buffer layer that improved charge collection and device stability [11]. In 2023, the same group further improved this efficiency, reaching 14.13 % thanks to advances in manufacturing processes and materials optimization [12]. The National Renewable Energy Laboratory (NREL) recently set a new world record for energy efficiency (PCE) for CZTSSe

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solar cells, achieving an optimum performance of 14.9 %. These results were found by the group headed by Cui [13]. Which is still well below the theoretical maximum efficiency of 32 % according to the Shockley-Queisser limit [14], and well below CIGS systems, which achieve an efficiency of 23.35 % [15]. This limitation in efficiency stems from three main factors. Firstly, the recombination of charge carriers in the material is induced by the presence of defects, as mentioned above [5,16]. Secondly, the junction between CdS and CZTSSe has physical incompatibilities, in particular an unfavorable energy alignment of the conduction bands (CBOs), which favors charge carrier recombination at this interface [17,18]. Finally, the junction between the CZTSSe and the molybdenum (Mo) electrode has surface defects that facilitate the dissipation of electrical charges [5]. Numerous studies have been carried out to reduce this recombination and optimize the performance of CZTSSe-based solar cells, because of these three mechanisms, which account for their low efficiency.

The search for substitute materials for the traditional CdS buffer layer is a major area of innovation in the development of CZTSSe photovoltaic cells. This approach meets several imperatives: optimizing optical performance, particularly at low wavelengths, improving the alignment of energy levels, and eliminating the health risks associated with cadmium [19,20]. Among the notable advances, ZTO (zinc-tin oxide) has stood out as a promising alternative. Ahmed et al. have demonstrated its potential by producing devices achieving an 11.8 % efficiency with an ultra-thin layer of ZTO (10 nm) [21]. Numerical simulations by Chargui and colleagues in 2024 even suggest the possibility of achieving higher performance, with theoretical efficiencies of 14.76 % for CZTS cells incorporating a ZTO buffer layer [5]. Another innovative approach, explored by Sun's team, is to use a $Zn_xCd_{1-x}S$ alloy whose composition can be modulated. Their work revealed that a specific proportion of zinc ($x = 0.35$) enabled yields over of 9 % to be achieved [22], paving the way for a significant reduction in cadmium content. The beneficial effect of these modifications is mainly explained by the optimization of the conduction band offset (CBO) resulting from the incorporation of zinc in the substitution of cadmium. A convincing illustration of this approach is provided by the work of Hao et al., who employed the SILAR method to deposit a buffer layer of $Zn_{0.35}Cd_{0.65}S$. This technique enabled optimum CBO to be obtained at the interface with the CZTS, leading to a remarkable yield of 9.2 % [22]. Recent advances in 2024 reflect innovative approaches, in particular those developed by Jingyuan Guo's team [19]. Their strategy, which consists of eliminating the traditional ZnO layer and incorporating indium ions into the CdS buffer layer, has led to a significant improvement in performance. This structural modification optimized the alignment of energy bands and minimized interfacial recombination losses, boosting the efficiency from 7.07 % to 9.01 %. In a complementary approach, Kim and colleagues explored the use of a nanoscale layer of In_2S_3 deposited on the CdS. This innovation had the effect of limiting charge carrier recombination and increasing the open circuit voltage (V_{oc}), a crucial parameter for the overall performance of the device [23]. In 2023, Changcheng Cui and his group improved the performance of CZTSSe solar cells by inserting a CZTSSe:In layer at the CdS/CZTSSe heterojunction interface using an In_2Se_3 -based post-deposition treatment. This method reduced Zn- and Sn-related deep defects, improved surface p-n conversion and increased the open circuit voltage from 488 mV to 527 mV, achieving an efficiency of 13.59 % [24]. In 2024, the same group continued these advances, achieving an efficiency of 14.5 % by combining low-temperature annealing and indium doping in the CdS buffer layer [25]. On the other hand, the WS_2 is emerging as a particularly interesting alternative in the field of photovoltaic components, particularly as a buffer layer [26–28]. This semiconductor has a range of physicochemical characteristics that set it apart from conventional materials. Apart from its direct gap and non-toxic nature, WS_2 is notable for its abundant natural availability. Its exceptional conductivity of around $10^3\Omega^{-1}cm^{-1}$ and high carrier mobility, which give it n-type behavior, make its electrical characteristics very alluring. It is a strong contender for optoelectronic applications because of these qualities [29,30]. Another significant benefit of this material is the variety of synthesis techniques. WS_2 thin films can be grown using a variety of deposition techniques, including CVD (chemical vapor deposition), CVT (chemical vapor transport), PVD (physical vapor deposition), ALD (atomic layer deposition), as well as sputtering [31,32]. This manufacturing flexibility, combined with its intrinsic thermal stability, makes it easy to integrate into industrial processes.

To solve the problem of carrier recombination at the interface between the absorber and the back electrode in solar cells, a back surface field (BSF) layer is often used to improve the efficiency of the device, particularly the open circuit voltage (V_{oc}) [33–35]. This approach, now established in various photovoltaic technologies, is based on the insertion of a specific intermediate layer that acts as a selective barrier. Its main role is to redirect minority carriers toward the depletion zone, thereby limiting recombination losses at the back interface [33–35]. The selection of the BSF material depends closely on the technology under consideration. High electrical conductivity, the capacity to create quasi-ohmic connections, and the best possible energy compatibility with the absorber layer are among the selection requirements. Several substances, such as Cu_2O [34], MoS_2, $MoSe_2$ [5], NiO [35], Sb_2S_3 [36], and CZTSSe [37], have shown promise. A significant breakthrough in this area was achieved by Chargui's team, who highlighted the exceptional performance of Cu_2O as a BSF layer in CZTS cells [5]. Their work demonstrated a spectacular improvement in efficiency, rising from 14.76 % to 23 %. The interest in Cu_2O extends beyond CZTS cells, also finding applications in CIGS and perovskite technologies, where it excels as a hole transport layer (HTL). Its success is due to an advantageous combination of optoelectronic properties, its non-toxicity, and its ease of processing.

This work is organized into three main parts. The first part focuses on the development of a numerical model for a CZTSSe solar cell, based on the SCAPS-1D simulator, enabling the accurate reproduction of experimental results for the Mo/CZTSSe/CdS/i:ZnO/Al:ZnO structure. The second section explores the potential of the WS_2 material as an alternative buffer layer to CdS, offering a solution to the toxicity problem associated with the latter. This section also examines the efficiency limitations of this solar cell through the analysis of carrier generation and recombination curves, and proposes an improvement through the addition of a BSF (Back Surface Field) layer. Finally, the third section assesses the impact of defects in the absorber layer and at the interfaces on photovoltaic performance. To complement this manuscript, an additional study is devoted to the influence of temperature on performance, including the extraction of key parameters such as activation energy and diffusion voltage, in order to further characterize the cell.

2. Structure and simulation methodology of the device

2.1. Device structure of the CZTSSe-based solar cell

In our research, we propose a solar cell using CZTSSe, designed in a similar way to traditional thin-film solar cells. We carried out a comparative analysis with a reference cell studied by Jiazheng Zhou and his team ("Fig. 1a" supplementary materials) [12]. Because of its high conductivity, transparency, and affordability, an Al:ZnO window layer is used in their design to facilitate the collection of produced electrons [38]. While the CdS buffer layer functions as an n-type semiconductor in the buffer structure, promoting the creation of the p-n junction and the accumulation of photogenerated charge carriers (electron-hole pairs) [33,39,40], an intrinsic layer of i:ZnO is also included to shield the CdS buffer layer during deposition. The absorber layer is composed of CZTSSe, a p-type semiconductor, while molybdenum (Mo) and aluminium (Al) are used for the back and front contacts, respectively.

We looked at the idea of substituting the non-toxic WS_2 buffer layer, which has demonstrated encouraging potential as a buffer layer in a

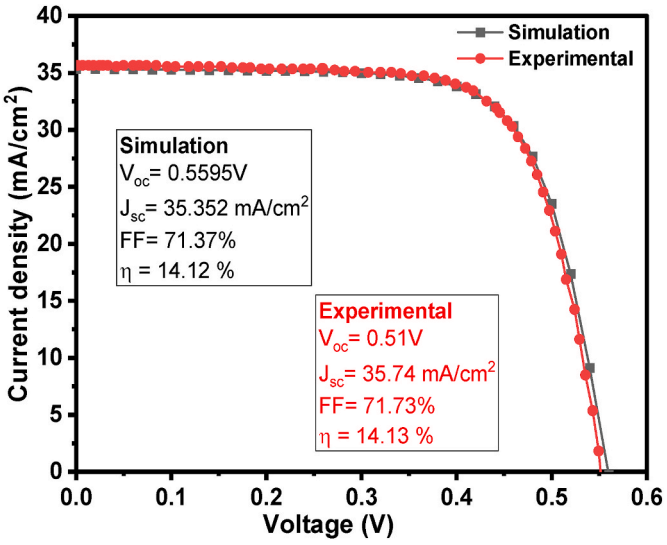


Fig. 1. Comparison between the J versus V characteristics for the experimental results [12], and the theoretical results found by our model.

variety of solar cell types, for the CdS buffer layer in order to solve the issue of CdS toxicity ("Fig. 1b" supplementary materials).

2.2. Numerical modeling and methodology for simulation

As part of this research, we assessed the suggested cell's performance using the SCAPS-1D simulator (version 3.3.09). One notable feature of this numerical tool is its one-dimensional approach to modeling photovoltaic behavior. It was created by Marc Burgelman's team at Ghent University in Belgium [41].

The mathematical foundation of SCAPS-1D is based on the simultaneous solution of three fundamental equations of semiconductor physics: the Poisson equation, the charge transport equations, and the continuity equations for electrons and holes [42,43]. SCAPS-1D is widely used to simulate organic [42], and inorganic solar cells [34], and can model cell structures with up to seven layers. It also offers the possibility of varying the simulation temperature and configuring the parameters of each layer. To achieve a realistic simulation, it is essential to take into account the different physical parameters of solar cells, obtained either experimentally or from theoretical calculations. Thanks to this numerical approach, SCAPS-1D calculates the electrical characteristics of the proposed solar cell, making it easier to optimize its performance and improve the device before any practical experimentation. For our simulation, the various physical parameters of the layers making up our cell are shown in Table 1. The simulation was carried out at a temperature of 300 K under standard A.M 1.5G solar lighting, with a power flux of 1000W/ cm^2.

3. Results and discussion

3.1. Validation of the numerical model for our CZTSSe-based solar cell through comparison with experimental results

To validate the reliability of our numerical approach, we carried out a meticulous calibration step by comparing the simulation results with the experimental data. Our approach involved the precise replication of the cell configuration analyzed by Jiazheng Zhou's team [12], which incorporated CdS as a buffer layer.

SCAPS-1D modeling was carried out under standard conditions, with the temperature set at 300K. The electrical structure of the contacts was faithfully reproduced, using molybdenum (Mo) for the rear contact and aluminium (Al) for the front contact, in line with the reference experimental set-up. All the physical parameters of each layer making up the cell were meticulously implemented in the simulator, in accordance with the data in Table 1. The specific characteristics of the metal contacts, detailed in Table 2, have also been taken into account to ensure exhaustive modeling of the device.

In order to confirm the effectiveness of our suggested numerical model, we introduce the notion of relative deviation. This measure quantifies the differences between the simulated results and the experimental results, in order to assess the accuracy of the numerical model. It is crucial for confirming whether the parameters and assumptions used in the simulation process accurately reflect the true behavior of the solar cell. The lower the value of δ, the more faithful the model is to the experimental results.

The following expression defines the relative deviation, denoted δ (equation (1)):

$$\delta = \frac{|Y_{sim} - Y_{exp}|}{Y_{exp}} \quad (1)$$

Where:

Y_{sim}: is the photovoltaic value obtained by simulation,

Table 2
Properties of solar cell contacts [5,35,47].

Contact parameters	Back contact (Mo)	Front contact (Al)
Work function	4.95	4.34
Surface Recombination Velocity of Electron (cm/s)	10^5	10^7
Surface Recombination Velocity of Hole (cm/s)	10^7	10^5

Table 1
Physical parameters of the layers in the simulated solar cell.

Physical parameters	Al:ZnO [5,44]	i:ZnO [5,44]	Absorber layer	Buffer layer		Back surface field (BSF)	
			CZTSSe [17,45]	CdS [35]	WS_2 [28,30]	Cu_2O [33,34]	V_2O_5 [46]
Thickness (μm)	0.5	0.1	1.1	0.05	0.05	0.03	0.03
Energy gap (eV)	3.3	3.3	1.146	2.45	2.1	2.17	2.2
Electron affinity (eV)	4.6	4.6	4.5	4.4	4.96	3.2	3.4
Relative permittivity	9	9	13.6	10	13.6	7.11	8
N_c effective density of states (cm^{-3})	2.2 x 10^{18}	2.2 x 10^{18}	2.2 x 10^{18}	2.2 x 10^{18}	2 x 10^{18}	2.02 x 10^{18}	9.2 x 10^{19}
N_v effective density of states (cm^{-3})	1.8 x 10^{19}	1.8 x 10^{19}	1.8 x 10^{19}	1.8 x 10^{19}	2 x 10^{19}	1.1 x 10^{19}	5 x 10^{20}
Electron mobility (cm^2/V. s)	100	100	100	100	100	200	150
Hole mobility (cm^2/V. s)	40	25	25	25	25	20	100
Shallow donor concentration N_D (cm^{-3})	1 x 10^{19}	1 x 10^{19}	0	1 x 10^{19}	1 x 10^{19}	0	0
Shallow acceptor concentration N_A (cm^{-3})	0	0	1 x 10^{18}	0	0	1. 10^{18}	1. 10^{18}
Electron/Hole thermal velocity (cm/s)	1 x 10^7	1 x 10^7	1 x 10^7	1 x 10^7	1 x 10^7	1 x 10^7	1 x 10^7
Defect type	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
Total density N_t (cm^{-2})	10^{14}	10^{14}	3 x 10^{16}	10^{14}	10^{14}	10^{14}	10^{14}

Y_{exp}: represents the photovoltaic quantity obtained experimentally.

The J-V curves obtained experimentally [12] and from our numerical model are shown in Fig. 1. The findings indicate that our model accurately reproduces the behavior of the solar cell, as demonstrated by the small value of δ, determined by the previous equation. This small deviation reflects a perfect balance between the experimental and simulation results. The corresponding values are shown in Table 3.

3.2. Optimizing CZTSSe solar cells with a WS_2 buffer layer and energy band alignment

After validating the numerical model, the CdS buffer layer in a CZTSSe-based solar cell was replaced with WS_2, a non-toxic alternative known for its efficiency in CIGS, Sb_2S_3, and perovskite solar cells. This modification retained the original structure except for the buffer layer. The resulting cell achieved notable performance, with an open-circuit voltage (V_{oc}) of 0.5613 V, a short-circuit current density (J_{sc}) of 35.64 mA/ cm^2, a fill factor (FF) of 64.27 %, and an efficiency of 12.8 %. Band diagrams for both CdS and WS_2 configurations are compared in Fig. 2b and c.

The term "Band shift" refers to the energy discontinuity at the interface between two semiconductors in a heterojunction, quantified by the conduction band offset (CBO = Δ E_c) and the valence band offset (VBO = Δ E_v). Equations (2) and (3) provide the means to calculate both types of offsets, Δ E_c and Δ E_v, which can be either positive or negative, as illustrated in Fig. 2b and c.

$$\mathrm{CBO} = \Delta E_c = \chi_{Absorber} - \chi_{Buffer} \tag{2}$$

$$VBO = \Delta E_v = \left(x + E_g\right)_{Absorber} - \left(x + E_g\right)_{Buffer} \tag{3}$$

A positive CBO indicates that the conduction band (CB) of the buffer layer is positioned higher than that of the absorber, forming a "spike" band profile that minimizes carrier recombination. Conversely, a negative CBO creates a "cliff" profile, leading to electron accumulation and increased recombination. At the CZTSSe/ WS_2 interface (Fig. 2c), the CBO is negative (−0.4 eV), resulting in a cliff that hampers performance. In contrast, the CZTSSe/CdS (Fig. 2b) interface exhibits a positive CBO (+0.1 eV), promoting better charge separation and reduced recombination. Furthermore, the VBO at the CZTSSe/CdS interface is 1.204 eV in conventional cells and increases to 1.41 eV in the proposed configuration. This enhancement ensures efficient hole retention at the interface, contributing to improved device efficiency.

In this work, we aim to determine the optimum operating conditions for the proposed solar cell. The thickness of the absorber layer and the doping density play a crucial role in solar cell performance. Optimum thickness is essential: too thin a layer limits light absorption, while too thick a layer, while it may increase current, may also reduce overall efficiency [48]. After validating the proposed Mo/CZTSSe/ WS_2/i: ZnO/Al:ZnO solar cell and replacing the toxic CdS with WS_2 as the buffer layer, efforts focused on optimizing the absorber layer. The WS_2 buffer layer's thickness was fixed at 0.05 μm according to the specifications in Table 1. While the CZTSSe absorber layer's thickness was varied from 0.2 μm to 3 μm, along with its doping density N_A varied from $10^{14} cm^{-3}$ to $10^{19} cm^{-3}$. Fig. 3 illustrates the effects of these adjustments on performance.

Table 3
Comparison of simulation results and experimental data, and the value of the observed variation (δ).

Research type	Open circuit voltage V_{oc} (V)	Short circuit current density J_{sc} (mA/ cm^2)	Fill factor FF(%)	Efficiency η (%)
Simulation	0.56	35.35	71.37	14.12
Experimental [12]	0.51	35.74	71.73	14.13
δ (%)	9.8	1.09	0.5	1.7

The results indicate that photovoltaic parameters such as open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}), Fill factor (FF) and efficiency (η) are strongly affected by variations in the thickness and doping density of the absorber layer. V_{oc} increases as a function of the doping density, independently of the layer thickness, due to better charge separation permitted by a higher doping density. J_{sc} depends mainly on the thickness of the absorber layer for thicknesses of less than 1 μm, increasing with thickness as it absorbs more photons. However, beyond 1 μm, J_{sc} becomes mainly influenced by the doping density (N_A), increasing with higher density thanks to improved charge carrier mobility [43].

The fill factor increases with higher doping density, irrespective of film thickness. This phenomenon can be explained by a drop in series resistance, encouraging more efficient transfer of charge into the cell [33]. In conclusion, we can see that the overall performance of the cell depends only on the density of the doping beyond 1 μm in thickness. It changes from 2.7 % in the case of a thickness of 1.2 μm and $N_A = 10^{15} cm^{-3}$ to 12.9 % in the case of the same thickness, but $N_A = 10^{18} cm^{-3}$. This large increase can be attributed to improved carrier density, which reduces recombination and optimizes charge collection. For optimum performance of our solar cell, we opt for a thickness of 1.1 μm and a doping density of $10^{18} cm^{-3}$ in the absorber layer.

A study of the effect of buffer layer thickness on the quantum efficiency of our solar cell shows that EQE improves with thicker WS_2 buffer layers in the short wavelength range (360–600 nm), but remains unchanged for longer wavelengths, as these photons are absorbed by the active layers. Detailed results are available in the supplementary materials associated with this manuscript.

The thickness of the buffer layer is also crucial to the solar performance of solar cells. Situated between the absorber layer and the transparent contact layer, this layer ensures that charges are merged and separated, thereby promoting the efficient collection of electrons and holes. It is therefore crucial to examine how its thickness affects the performance of the solar cell. In Fig. 4, we examine how the thickness of the WS_2 buffer layer, which varies between 0.05 μm and 1 μm, and that of the absorber layer, which fluctuates between 0.2 μm and 3 μm, influence the performance of the solar cell. The results in Fig. 4 show that the efficiency of the solar cell is almost independent of the thickness of the buffer for a thickness of less than 1 μm, while for a thickness greater than 1 μm, there is an increase in efficiency as the thickness of the WS_2 increases. Based on these values, we opt for an optimum WS_2 thickness of 0.1 μm, while the doping density is fixed at $10^{19} cm^{-3}$, as demonstrated by previous research carried out by our team, which indicates that the doping density has no influence on the photovoltaic parameters of the solar cell.

3.3. Effect of back surface field

The essential role of the BSF (Back Surface Field) layer is to act as a selective barrier for charge carriers. On the one hand, it produces an electric field that moves minority carriers away from the back surface, thereby reducing recombination at the contact between the absorber and back contact [34,35]. On the other hand, it facilitates the movement of predominant carriers towards the back metal contact by reducing contact resistance, which optimizes current collection. With these functions doubled, the BSF layer plays a crucial role in improving the efficiency of the solar cell. According to the results obtained, we note a low efficiency of our cell thanks to the use of WS_2 as a buffer layer. To optimize the efficiency of our cell and reduce charge carrier recombination, we use a layer that acts as a BSF. In our research, we are comparing two layers used as BSFs: Cu_2O and V_2O_5 ("Fig. 1c" supplementary materials). To ascertain these parameters (thickness and doping density), we adjusted the thickness of the back surface field (BSF) layer within the range of 0.01 μm–0.2 μm, while the doping density was

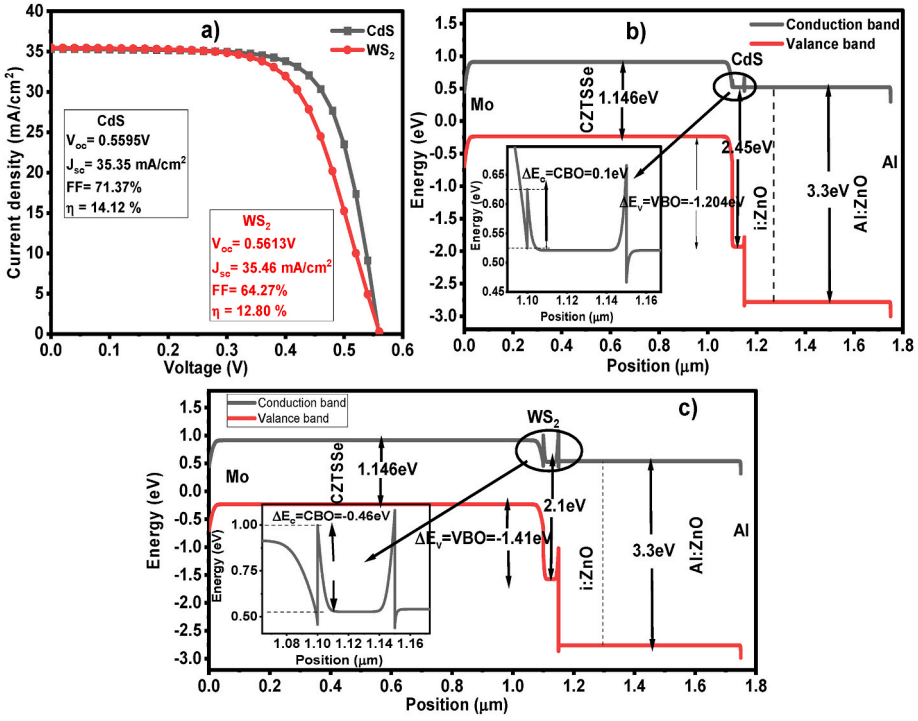


Fig. 2. a) J-V characteristics with WS_2 and CdS as buffer layers; b) Energy band diagram using CdS as a buffer layer; c) Energy band diagram using WS_2 as a buffer layer.

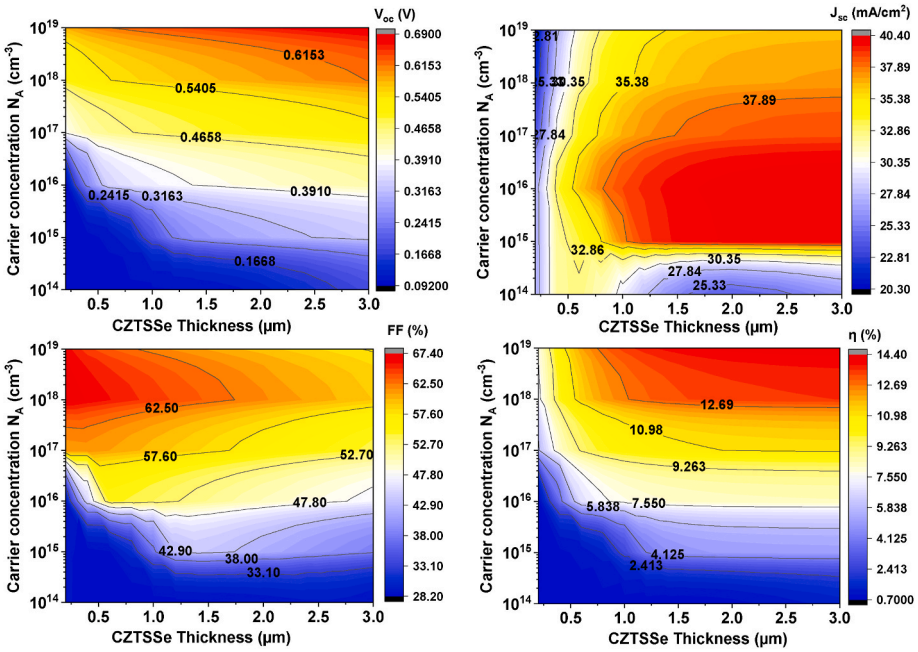


Fig. 3. Effect of varying doping density and absorber layer thickness in the Mo/CZTSSe/ WS_2/i:ZnO/Al:ZnO structure on the performance of the proposed solar cell.

varied from $10^{14}cm^{-3}$ to $10^{19}cm^{-3}$. The effects of these modifications are illustrated in Fig. 5.

Fig. 5b illustrates that the thickness of the BSF layer, whether composed of Cu_2O or V_2O_5, does not significantly alter the photovoltaic parameters. Furthermore, doping density (Fig. 5a) does not notably affect these parameters as long as it remains below $10^{17}cm^{-3}$. However, when the doping density exceeds this threshold, a slight enhancement in photovoltaic performance is observed due to the incorporation of the two BSF layers. This improvement aligns with findings reported in the literature, where BSF layers have been shown to enhance carrier collection and reduce recombination. The BSF layer acts as a barrier that prevents carrier recombination at the interface between the absorber and the rear metal contact, while generating an electric field that drives minority carriers away from the rear surface, ultimately improving charge transport and collection efficiency.

This effect is clearly demonstrated in the band diagram in Fig. 6a, which reveals the formation of an energy barrier at the CZTSSe/Mo interface, limiting recombination. The inclusion of a Back Surface Field

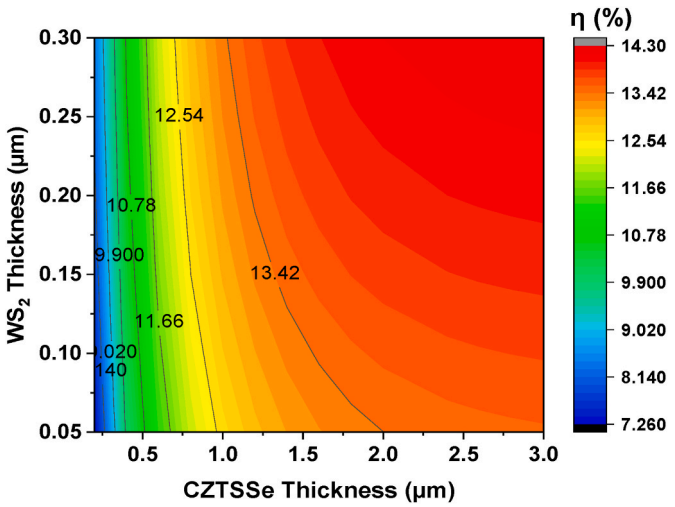


Fig. 4. Impact of varying the thickness of the absorber layer and buffer layer on solar cell efficiency.

(BSF) layer plays a crucial role in reducing the loss of photogenerated carriers, particularly minority carriers, thereby improving carrier extraction and enhancing overall device performance. Recent research has shown that the rear interface is influenced not only by carrier recombination but also by various factors such as element inhomogeneity and heterogeneous nucleation [13,49,50].

Fig. 6b presents the J-V curves of the proposed solar cell with and without the BSF layer. The results indicate a significant improvement in photovoltaic parameters when the BSF layer is incorporated. In the absence of a BSF and using WS_2 as a buffer layer, the solar cell performance reaches a $V_{oc} = 0.56$ V, $J_{sc} = 35.46$ mA/ cm^2, a fill factor (FF) of 64.27 %, and an efficiency of 12.8 %. By incorporating Cu_2O as a BSF layer, performance increases to $V_{oc} = 0.72$ V, $J_{sc} = 38.99$ mA/ cm^2, a FF of 73.42 %, and an efficiency of 19.77 %. With V_2O_5 as the BSF layer, performance reaches $V_{oc} = 0.73$ V, $J_{sc} = 38.10$ mA/ cm^2, an FF of 73.42 %, and an efficiency of 19.07 %. These results strongly support the positive impact of the BSF layer on the solar cell's efficiency.

Additionally, the efficiency improvements observed with Cu_2O or V_2O_5 as BSF materials suggest that these materials may play a significant role in enhancing charge collection and minimizing recombination. Cu_2O, with its suitable band alignment and potential for efficient charge separation, and V_2O_5, known for its high conductivity and stability, both contribute to the reduction of energy loss mechanisms at the rear interface. This enhancement is crucial for future developments in high-efficiency thin-film solar cells, where optimizing the rear contact and minimizing recombination are key factors in improving overall performance. Further investigations into the material properties and interface engineering of the BSF layer could potentially lead to even higher efficiencies, and for other applications [51–53]

3.4. Generation and recombination rate profile

It is essential to analyze the "generation and recombination rate" profile present in thin solar cells in order to understand their capacity to convert light energy into electricity. The generation rate indicates where photons are absorbed and electron-hole pairs are formed. An ideal generation profile is found near the p-n junction or in the main absorption layer, guaranteeing efficient energy conversion. At the same time, the degree of recombination indicates the regions where charge carriers degrade before being reunited, reducing the overall efficiency of the cell. Recombination is generally concentrated near interfaces or imperfections in thin films, where losses can occur.

We have drawn the recombination rate profile for the cell in the presence and absence of the BSF layer (Cu_2O or V_2O_5) for our suggested solar cell, employing WS_2 as the buffer layer, as shown in Fig. 7. The charge carrier production curve upon absorption of photons from the solar spectrum is clearly illustrated in Fig. 7a. In the absence of BSF, it is observed that this layer does not generate any charge carriers, which confirms previous results showing that the thickness of the BSF does not directly influence the cell's performance. Furthermore, the generation of charge carriers increases with the thickness of the absorber layer, due to the increased absorption of the solar spectrum. There is also a

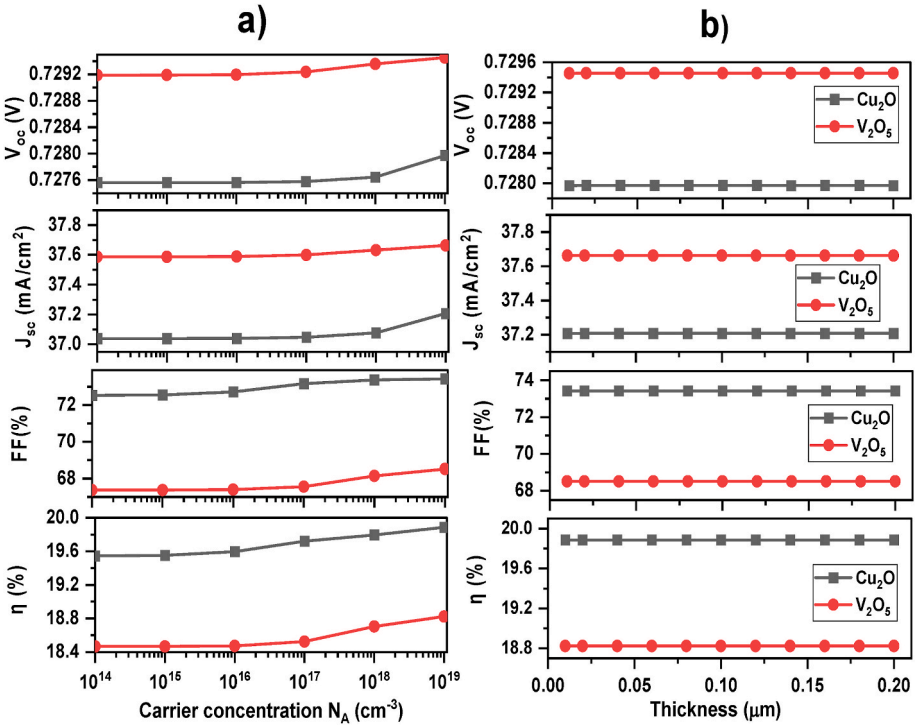


Fig. 5. Impact of a) Thickness and b) Doping density (N_A) of the BSF layer on solar cell performance.

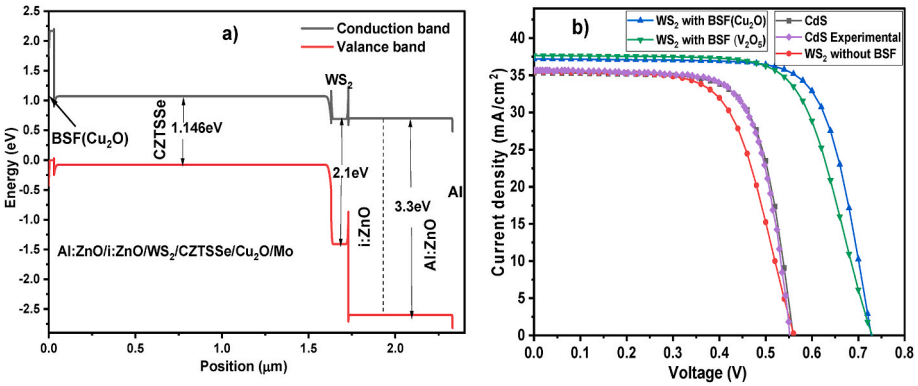


Fig. 6. a) Energy band diagram of the proposed solar cell with back surface field (BSF); b) J-V characteristics of the experimental solar cell with and without BSF.

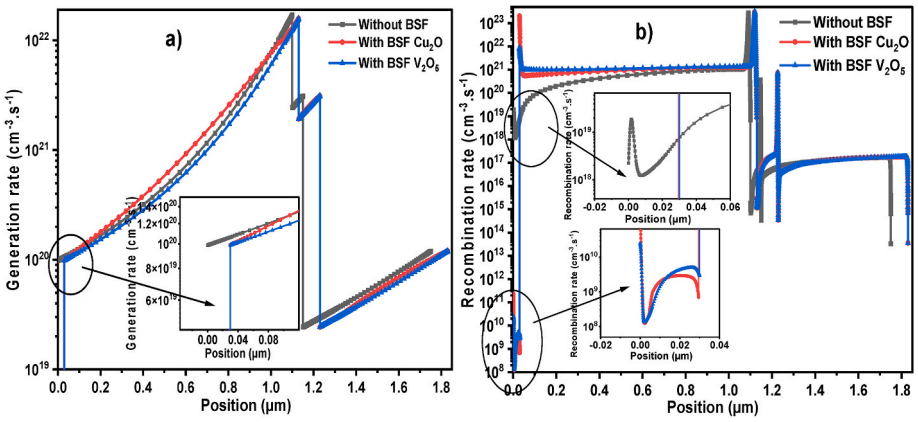


Fig. 7. a) Generation rate and b) Recombination rate of the proposed solar cell with and without the back surface field (BSF) Layer.

discontinuity in charge carrier generation at the interface between the absorber layer and the buffer layer, indicating that the energy gap of the buffer layer is not optimal for absorbing photons from the solar spectrum.

As for the recombination curve Fig. 7b, the results confirm the role of the BSF layer in reducing recombination at the interfaces. Without BSF, the charge carrier recombination rate at the absorber/back contact metal (Mo) interface is of the order of $10^{18} cm^{-3}.s^{-1}$. In the presence of BSF, this rate decreases to about $10^{8} cm^{-3}.s^{-1}$, as shown in the inset of Fig. 7b, demonstrating that BSF effectively reduces charge carrier recombination at the CZTSSe/Mo interface. These results are also in agreement with existing studies that show an increase in the recombination rate at the contact interfaces between the different layers, due to the appearance of energy bands that trap charge carriers, thus increasing recombination at these interfaces.

3.5. Effect of bulk defect density (N_t) of CZTSSe layer

One of the major causes of charge carrier recombination is the presence of defects in the absorber layer. This results from the absorption of photons and affects the overall performance of the solar cell, hindering the achievement of the theoretical Shockley-Queisser threshold. These anomalies act as traps or recombination spaces for charge carriers. During their production, it is necessary to improve the quality of the absorbing film to reduce the presence of defects. The density of defects N_t in the absorber layer is directly related to the Shockley-Read-Hall recombination rate (R^{SRH}), which is expressed by equation (4) [54]:

$$R^{SRH} = \sigma_{n,p}.\ V_{th}.\ N_t \left[\frac{n.p - n_i^2}{n + p + 2n_i \cos\left(\frac{E_t - E_i}{K_B T}\right)} \right] \tag{4}$$

Where σ_n and σ_p represent the electron and hole capture cross sections, respectively, V_{th} the thermal velocity, N_t the defect density, E_t the energy level of the trap states, and E_i the intrinsic energy level.

To improve the collection of charge carriers, it is essential to reduce their diffusion length L, which is given by equation (5) [54]:

$$L_{n,p} = \sqrt{D_{n,p}\tau_{n,p}} \tag{5}$$

With $D_{n,p}$ representing the diffusion coefficient, determined by equation (6) [54]:

$$D_{n,p} = \frac{K_B T}{q}\ \mu_{n,p} \tag{6}$$

And $\tau_{n,p}$ the lifetime of the photogenerated charge carriers, defined by equation (7) [54]:

$$\tau_{n,p} = \frac{1}{\sigma_{n,p}\ N_t\ V_{th}} \tag{7}$$

The effect of total defect density N_t on solar cell performance is shown in Fig. 8. A high defect density decreases charge carrier longevity, leading to increased recombination and reduced diffusion distance. Consequently, a high defect density promotes carrier recombination, thereby reducing the efficiency of the solar cell.

To evaluate the effect of defect density in the absorber layer on the performance of our proposed solar cell, we varied N_t from $10^{14} cm^{-3}$ to

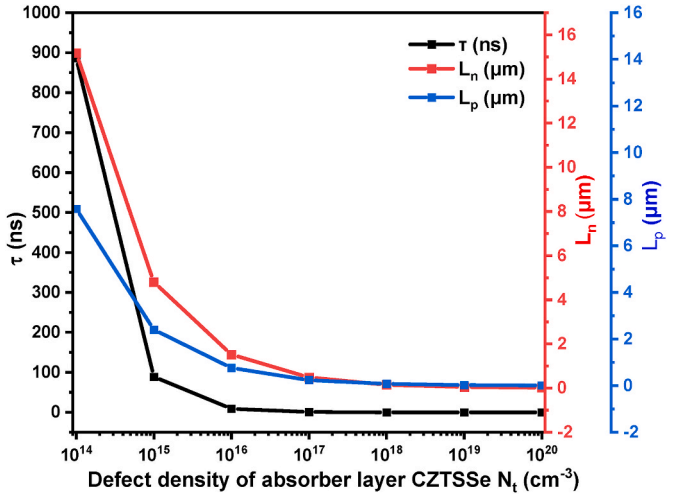


Fig. 8. Influence of defect density on charge carrier diffusion length and lifetime.

$10^{20}cm^{-3}$, both in the absence and presence of the back field layer (BSF). The results illustrated in Fig. 9 show a decrease in all photovoltaic parameters as the defect density increases, whether the BSF layer is present or not. For example, the open-circuit voltage V_{oc} drops from 0.56 V for a defect density of $10^{14}cm^{-3}$ to 0.28 V for $10^{20}cm^{-3}$ in the absence of BSF. In contrast, in the presence of BSF, particularly with Cu_2O, V_{oc} drops from 0.89 V for $N_t = 10^{14}cm^{-3}$ to 0.31 V for $N_t = 10^{20}cm^{-3}$. Similar behaviour is observed when V_2O_5, is used as BSF. This decrease in performance is attributed to a reduction in the lifetime of the charge carriers generated by photon absorption, as well as a decrease in the carrier scattering length, as discussed previously.

To illustrate the effect of defect density in the absorber layer on the J-V characteristics of our proposed solar cell in the absence of BSF, we present the graph in Fig. 10. This plot highlights that increasing the defect density (N_t) in the CZTSSe absorber layer results in a significant degradation of the solar cell performance. At low defect density

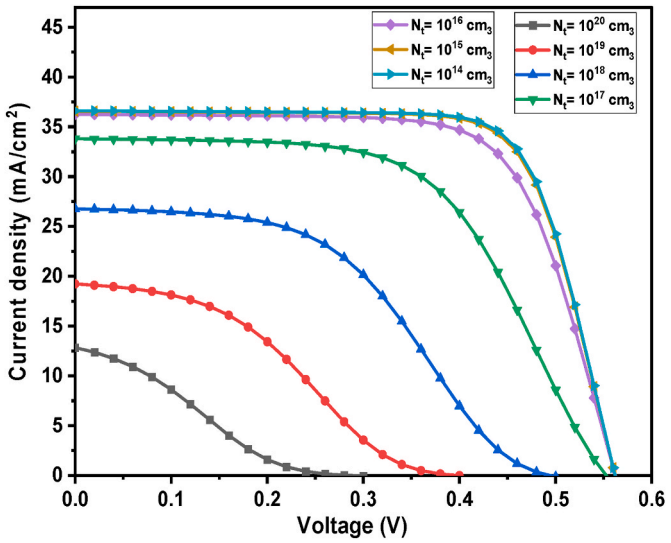


Fig. 10. Impact of defect density in the absorber layer of the cell without BSF on J-V characteristics.

($10^{14}cm^{-3}$), the cell exhibits a high open-circuit voltage (V_{oc}) of around 0.55 V and a short-circuit current density (J_{sc}) of around 40mA/ cm^2. However, when N_t reaches $10^{20}cm^{-3}$, V_{oc} falls to around 0.1 V and J_{sc} decreases significantly, demonstrating that the defect density strongly alters the cell's ability to generate current.

3.6. *Impact of interface defects at WS_2/CZTSSe and CZTSSe/BSF junctions*

The power conversion efficiency (PCE) of a solar array depends on the characteristics of the interface layer. A high concentration of defects in these layers compromises their quality by increasing obstacles to charge carriers, leading to additional recombination losses and, ultimately, reducing the overall efficiency of the system. The following equation (8) describes the recombination process that occurs between

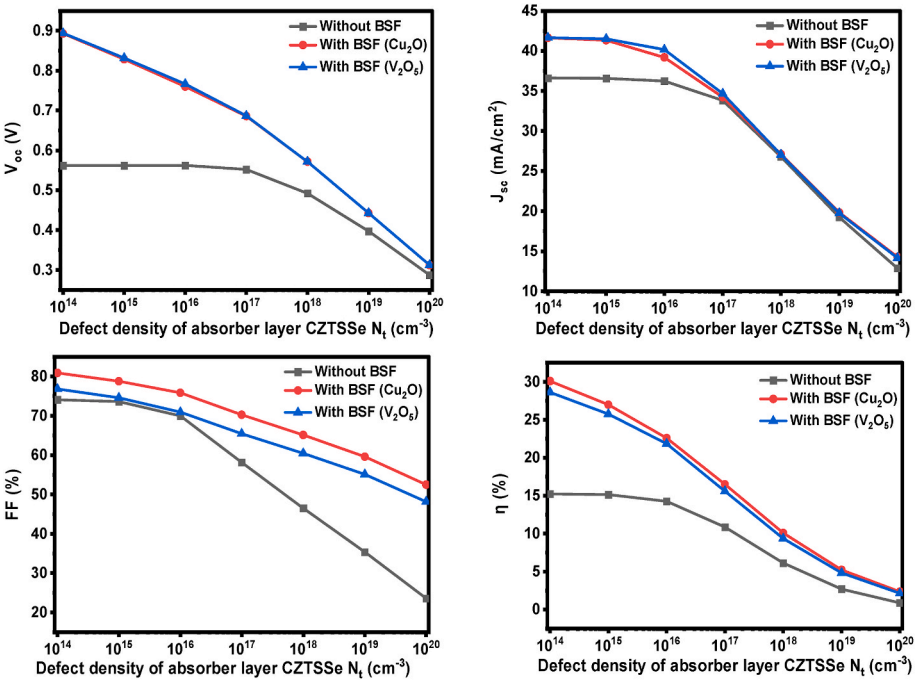


Fig. 9. Impact of absorber layer (CZTSSe) defect density on the parameters of the proposed photovoltaic cell with and without the back surface field (BSF) layer.

the absorber and the interface layer [55]:

$$R_{interface} = \left[\frac{p_{interfce} \cdot n_{interface} - n_i^2}{S_p^{-1}\left(n_{interface} + n\right) + S_n^{-1}\left(p_{interface} + p\right)} \right] \quad (8)$$

In this formula $p_{interface}$ and $n_{interface}$ symbolise respectively the hole and electron concentrations at the interface, while n_i corresponds to the intrinsic carrier concentration. The rate of recombination of holes and electrons, called S_p and S_n, is influenced by the defect density ($N_{t(iF)}$), the thermal rate ($V_{th(n,p)}$) and the effective capture segment ($\sigma_{(n,p)}$), as shown in equation (9) below:

$$S_{(n,p)} = N_{t(iF)} V_{th(n,p)}\ \sigma_{(n,p)} \quad (9)$$

In this study, we investigated the impact of defect density in the interface layer on the performance of the solar cell. As outlined in Table 4, we varied the defect density (N_t) at the CZTSSe/ WS_2 and BSF/CZTSSe interfaces by introducing interface defects. As illustrated in Fig. 11, the total defect density was adjusted from $10^{10} cm^{-3}$ to $10^{16} cm^{-3}$ for the WS_2/CZTSSe and CZTSSe/BSF interfaces.

The results indicate a strong correlation between interface defect density and solar cell performance. At lower defect densities, the efficiencies improve, reaching 19.69 % for the Cu_2O BSF and 19.09 % for V_2O_5. This enhancement can be attributed to reduced carrier recombination at the interfaces, allowing for better charge separation and collection. However, as defect density increases, efficiency drops significantly, which can be explained by the corresponding increase in charge carrier recombination rates at the interfaces [5,30,32]. This decrease in efficiency with higher defect densities highlights the crucial role of interface quality in determining the overall performance of thin-film solar cells. Minimizing defect density at these interfaces is essential to reduce recombination losses and maximize carrier lifetime, thus improving device efficiency. These findings emphasize the importance of precise control over interface engineering in the development of high-efficiency solar cells. Further optimization of defect management at the CZTSSe/ WS_2 and BSF/CZTSSe interfaces could lead to significant improvements in solar cell performance.

The effect of temperature on the proposed CZTSSe-based solar cell, with the use of WS_2 as a buffer layer and BSF, is studied in detail in the supplementary materials associated with this manuscript. Using the relationship between the evolution of open-circuit voltage and temperature, the activation energy (E_a) can be determined, which is a key parameter influencing solar cell performance, particularly with regard to recombination centers at the buffer-absorber interface and in the depletion region. These recombination centers are due to inconsistencies in energy levels between the different layers. The activation energy can be obtained from the relationship between V_{oc} and temperature, as shown in equation (10). Recombination losses in thin-film solar cells occur in several regions: the space charge region (SCR), as well as in the front, bulk and back sections of the cell. This includes bulk recombination, interface losses and contact recombination, all of which affect overall efficiency.

Table 4
Physical properties of the simulated interface.

Parameters	CZTSSe/W S_2	Cu_2 O/CZTSSe	V_2O_5/CZTSSe
Defect type	Neutral	Neutral	Neutral
Electron Capture cross-section (cm^2)	10^{-16}	10^{-16}	10^{-16}
Hole Capture cross-section (cm^2)	10^{-16}	10^{-16}	10^{-16}
Defect energy level (E_t) reference	Above the highest E_v	Above the highest E_v	Above the highest E_v
Energy with respect to reference (eV)	0.6	0.6	0.6
Total density (cm^{-3})	$10^{10} - 10^{16}$	$10^{10} - 10^{16}$	$10^{10} - 10^{16}$

$$V_{oc} = \frac{E_a}{q} - A\ K_B\ T\ q \ln\left(\frac{J_0}{J_L}\right) \quad (10)$$

In thin-film solar cells (TFSCs), the activated recombination energy can be used to assess the main losses. Losses due to Shockley-Read-Hall (SRH) recombination have been included in the model. Interfacial recombination between the absorber layer and the buffer layer is often the main recombination mechanism in solar cells when $E_a < E_{g\ Absorber}$, as previous research indicates. In contrast, the main recombination process is usually SRH recombination in the space charge region (SCR), which predominates at the interface when $E_a = E_{g\ Absorber}$ or $E_a > E_{g\ Absorber}$.

Interfacial recombination in TFSCs occurs when electrons and holes recombine at the interfaces of the different device layers. In our study, we found that $E_a = 1.2714$ eV for Cu_2O and $E_a = 1.2816$ eV for V_2O_5 (see supplementary materials.) These values are significantly higher than the gap energy of the CZTSSe absorber layer, which is 1.14 eV. Consequently, we can conclude that, for both devices studied, the dominant recombination is of the SRH type in the space charge region (SCR) of the absorber layer, which is also confirmed by the results presented in Fig. 12, illustrating the J_{SRH}-V curves.

This analysis shows the importance of recombination management at the interface and in the space charge region for optimizing the efficiency of CZTSSe-based solar cells. The results also highlight the importance of interface design and activation energy control to minimize recombination losses and improve overall device performance.

In supplementary materials, we analyze the C-V and C-f curves of our solar cell equipped with Cu_2O and V_2O_5 back surface field (BSF) layers. The C-V curves are used to determine the doping profile, depletion zone and diffusion potential (V_d), while the C-f curves provide information on deep defects, trap reaction times and charge carrier behaviour, as well as material insertion and extraction processes. These analyses provide valuable insights into the operation and performance of the solar cell.

4. Conclusion

This study validated a numerical model of a CZTSSe thin-film solar cell (Mo/CZTSSe/CdS/i:ZnO/Al:ZnO structure) using the SCAPS-1D simulator, obtaining simulation results in agreement with experimental data in the literature. We demonstrated that the WS_2 material, used as a buffer layer, can advantageously replace CdS, eliminating the toxicity issue while maintaining good performance. Optimization of the CZTSSe absorber layer enabled us to achieve an efficiency of 12.80 % with a thickness of 1.1 µm and a doping density of 10^{18} cm^{-3}, albeit limited by charge carrier recombination at the interfaces. To overcome these losses, the introduction of a BSF (Back Surface Field) layer between the CZTSSe and the metal contact was proposed. A comparative analysis between Cu_2O and V_2O_5 as BSF materials revealed a significant improvement in efficiency, reaching 19.77 % and 19.07 % respectively. This improvement is attributed to a reduction in carrier recombination, as confirmed by generation and recombination analyses. The study also explored the effect of defects in the absorber layer and at the interfaces, as well as the influence of temperature on cell performance. Key parameters, such as the activation energy E_a (1.2714 eV for Cu_2O and 1.2816 eV for V_2O_5) and diffusion voltage values V_d of 1.4 V and 1.54 V for V_2O_5 respectively, as well as the C-V characteristics, were analyzed, providing essential information for understanding the recombination mechanisms. These results represent a breakthrough in the optimization of CZTSSe solar cells, offering prospects for more efficient and environmentally friendly photovoltaic devices.

CRediT authorship contribution statement

Taoufik Chargui: Writing – original draft, Software, Methodology, Conceptualization. **Fatima Lmai:** Writing – review & editing, Validation, Supervision. **Khalid Rahmani:** Writing – review & editing,

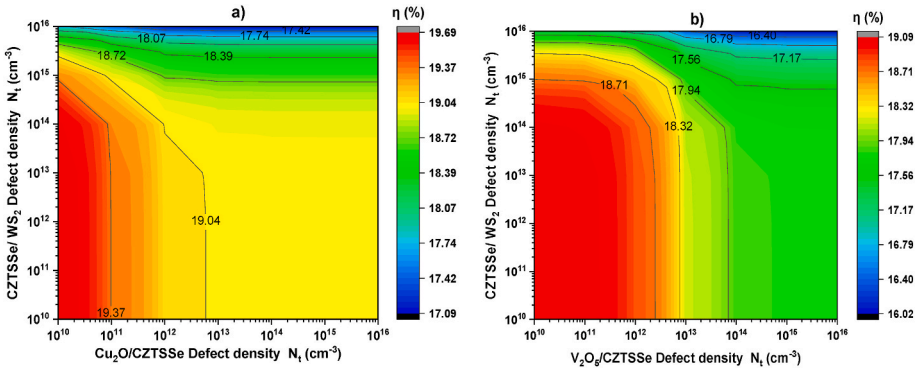


Fig. 11. Impact of defect density of the BSF/CZTSSe and CZTSSe/ WS_2 interface on solar cell efficiency.

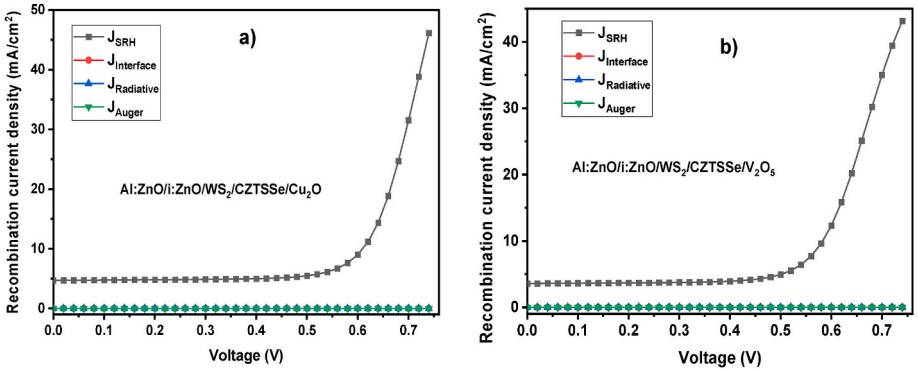


Fig. 12. Current density recombination a) for Cu_2O as BSF b) V_2O_5 as BSF.

Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.solmat.2025.113424.

Data availability

Data will be made available on request.

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